

Nathen Paniagua

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EDUCATION

California State University, Fullerton

Aug 2023 - May 2027

Bachelor of Science, Computer Science

- **Achievements:** Expected Graduation - May 2027
- **Coursework:** Object Oriented Programming, Data Structures, Databases & File Systems, Software Engineering, Compilers & Languages, Data Science & Big Data

TECHNICAL SKILLS

- **Programming Languages:** C++, Python, Swift, JavaScript, R
- **Frameworks:** React, Node.js, Tailwind CSS

RESEARCH EXPERIENCE

Student Researcher – G π T: Generative Practice Interview Trainer

Jan 2026 - Present

- Conduct research on AI-driven interview systems analyzing student interaction data, including transcripts and survey responses
- Apply qualitative and quantitative data analysis techniques to identify patterns in user performance and feedback
- Contribute to iterative development and evaluation of an AI-based educational tool aimed at enhancing student interview preparedness
- Assist in preparing research findings for presentations, reports, and potential conference submissions

PROJECTS

Nora - On-Device Pre-Triage Co-Pilot

Apr 2026 - Apr 2026

- Architected a privacy-first Android triage app in Kotlin + Jetpack Compose that orchestrates three on-device ML models (Qwen 3.5B LLM, Whisper Tiny ASR, CLIP ViT-Base) via the Melange SDK, routing patients to one of four care tiers in under 6 seconds with zero PHI leaving the device.
- Engineered a multi-layer inference pipeline combining a deterministic regex emergency short-circuit, CLIP zero-shot medical image classification, and an LLM triage engine with prompt prefill forcing and 160-token budget caps to achieve structured JSON output on memory-constrained mobile hardware.
- Implemented a bidirectional PHI de-identification system using a regex anonymizer and encrypted token map (Android Keystore), enabling safe transmission of de-identified documents to telehealth providers with full re-identification post-response.
- Designed a FakeMelangeRuntime abstraction layer enabling parallel UI and ML development with no shared dependencies, reducing integration friction and maintaining a working demo build throughout the hackathon.

Gemini-Integrated Smart Fridge Assistant

Aug 2025 - Dec 2025

- Developed a full-stack iOS application utilizing SwiftUI for a dynamic user interface and Core Data for efficient, local object persistence (fridge inventory), ensuring seamless offline data management.
- Engineered a secure, asynchronous API service layer connecting the mobile frontend to a serverless Firebase Cloud Functions (Node.js) backend, enabling real-time, personalized recipe generation via the Gemini API.
- Collaborated effectively with a three-person team throughout the SDLC (Software Development Lifecycle), successfully bridging mobile and cloud environments to deliver a scalable, multi-component architecture.
- Implemented robust data transformation and mapping logic to convert unstructured large language model (LLM) text responses into a structured JSON format, guaranteeing a clean and accurate display of recipes and user profile settings.

LEADERSHIP & ORGANIZATIONS

CSUF Theta Tau Professional Engineering Fraternity | *Potential New Member Parent*

Aug 2025 - Dec 2025

- Directed mentorship and onboarding for 20+ engineering students, leading weekly development sessions focused on academic success, teamwork, and professional readiness, achieving a 100% program completion rate.
- Served as the communication bridge between executive leadership and new members, facilitating information flow and conflict resolution that maintained organizational transparency and member satisfaction above 90%.
- Coordinated interdisciplinary team projects blending mechanical engineering, computer science, computer engineering, and electrical engineering principles, fostering collaboration and technical confidence among new members.
- Applied data-driven feedback loops from surveys and one-on-one check-ins to identify skill gaps, iteratively improving mentorship materials and boosting overall participant engagement by 30%.